

Task 2 - Network Security & Scanning

NMAP NETWORK SCAN REPORT

Cybersecurity Internship – Practical Assessment

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Operating System: Kali Linux

Target: Metasploitable2 – Controlled Lab Environment **Tool:**
Nmap

TCP SYN Scan • UDP Scan • Service Version Detection • OS Detection

Table of Contents

1. Introduction
2. Objective
3. Lab Environment
4. Methodology
5. TCP SYN Scan (-sS)
6. UDP Scan (-sU)
7. Service Version Detection (-sV)
8. OS Detection (-O)
9. Findings and Security Analysis
10. Recommendations
11. Conclusion

1. Introduction

Nmap (Network Mapper) is a network discovery and security auditing tool used to identify hosts, open ports, services, service versions, and operating-system information. This practical was performed in an authorized virtual laboratory using Kali Linux and Metasploitable2.

2. Objective

The objectives were to perform TCP and UDP scans, identify services and versions, attempt OS detection, interpret the results, and document security-relevant findings.

3. Lab Environment

Scanning system: Kali Linux

Target: Metasploitable2

Environment: Private VirtualBox laboratory network

Tool: Nmap

4. Methodology

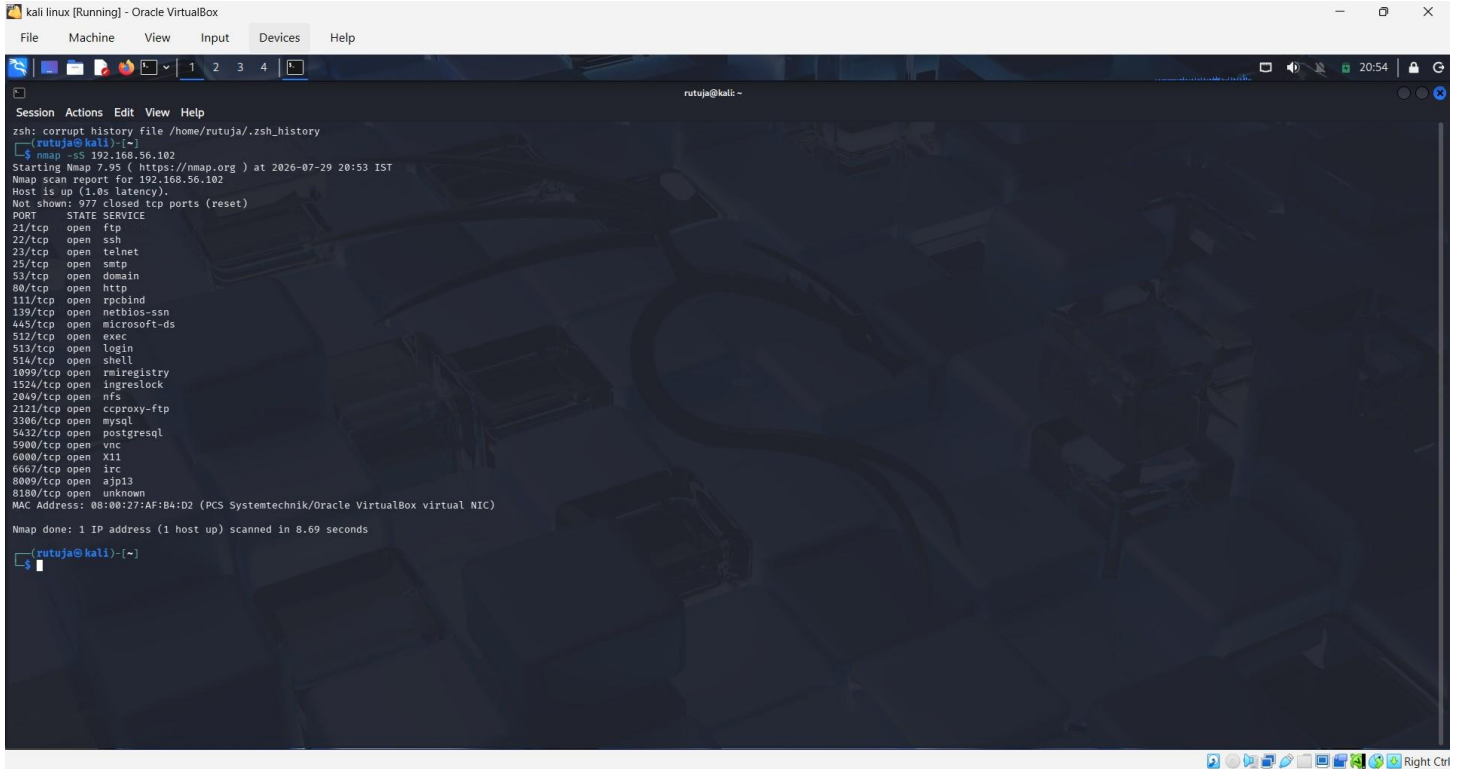
The following Nmap techniques were used.

Technique	Option	Purpose
TCP SYN	-sS	Discover TCP ports
UDP	-sU	Discover UDP services
Service detection	-sV	Identify services and versions
OS detection	-O	Estimate target operating system

5. TCP SYN Scan (-sS)

A TCP SYN scan sends SYN probes to TCP ports and analyzes the responses to determine whether ports are open, closed, or filtered. It is commonly used for efficient TCP port discovery.

Command: `nmap -sS 192.168.56.102`



```
kali linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help

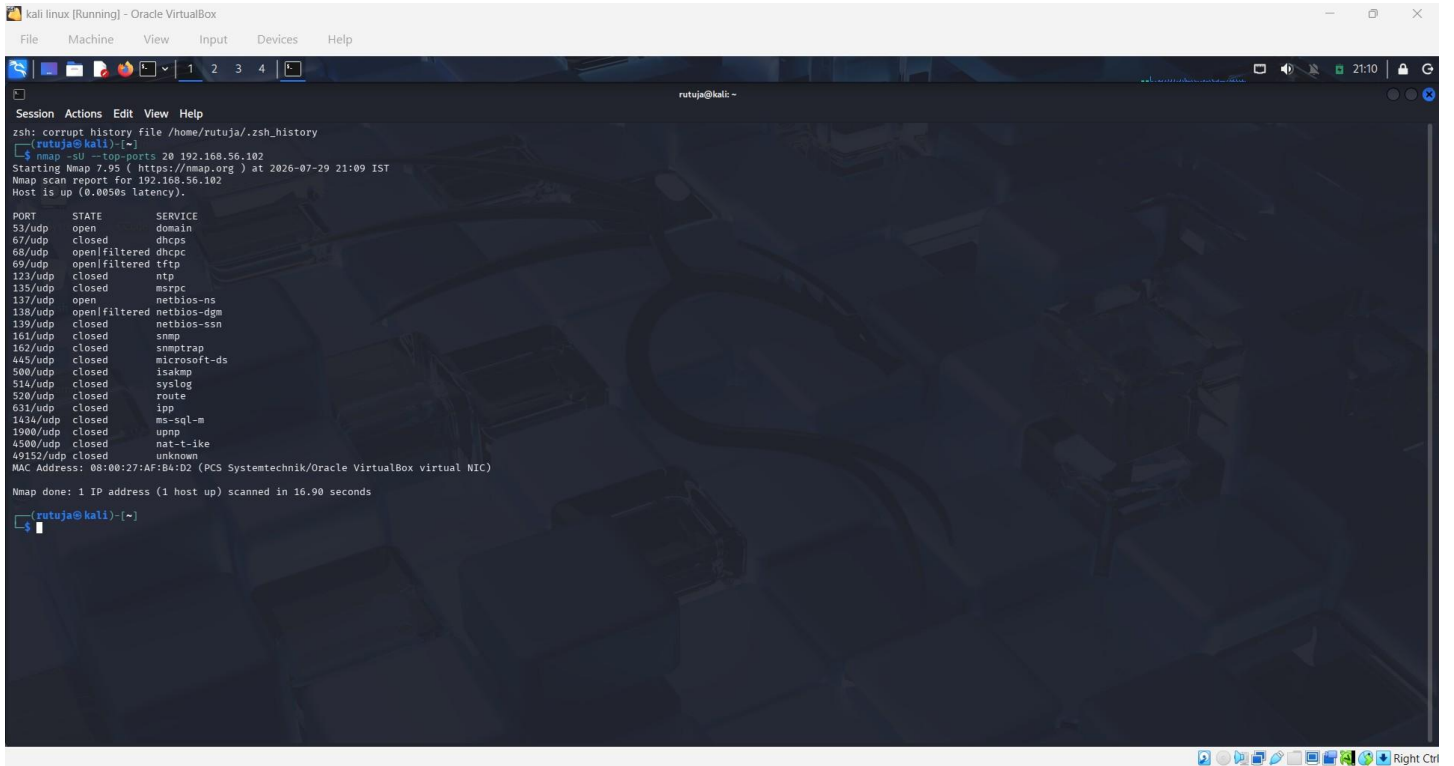
rutuja@kali: ~
zsh: corrupt history file /home/rutuja/.zsh_history
[rutuja@kali]~$ nmap -sS 192.168.56.102
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-29 20:53 IST
Nmap scan report for 192.168.56.102
Host is up (1.8s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmlregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
3432/tcp  open  postgresql
3900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:AF:B4:D2 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 8.69 seconds
[rutuja@kali]~$
```

6. UDP Scan (-sU)

A UDP scan checks UDP ports. Because UDP does not use a TCP connection handshake, Nmap uses responses and other indicators to classify ports. UDP scans can take longer than TCP scans.

Command: `nmap -sU --top-ports 20 192.168.56.102`



```
kali linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help

rutuja@kali: ~
Session Actions Edit View Help
zsh: corrupt history file /home/rutuja/.zsh_history
rutuja@kali:~$ nmap -sU --top-ports 20 192.168.56.102
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-29 21:09 IST
Nmap scan report for 192.168.56.102
Host is up (0.0050s latency).

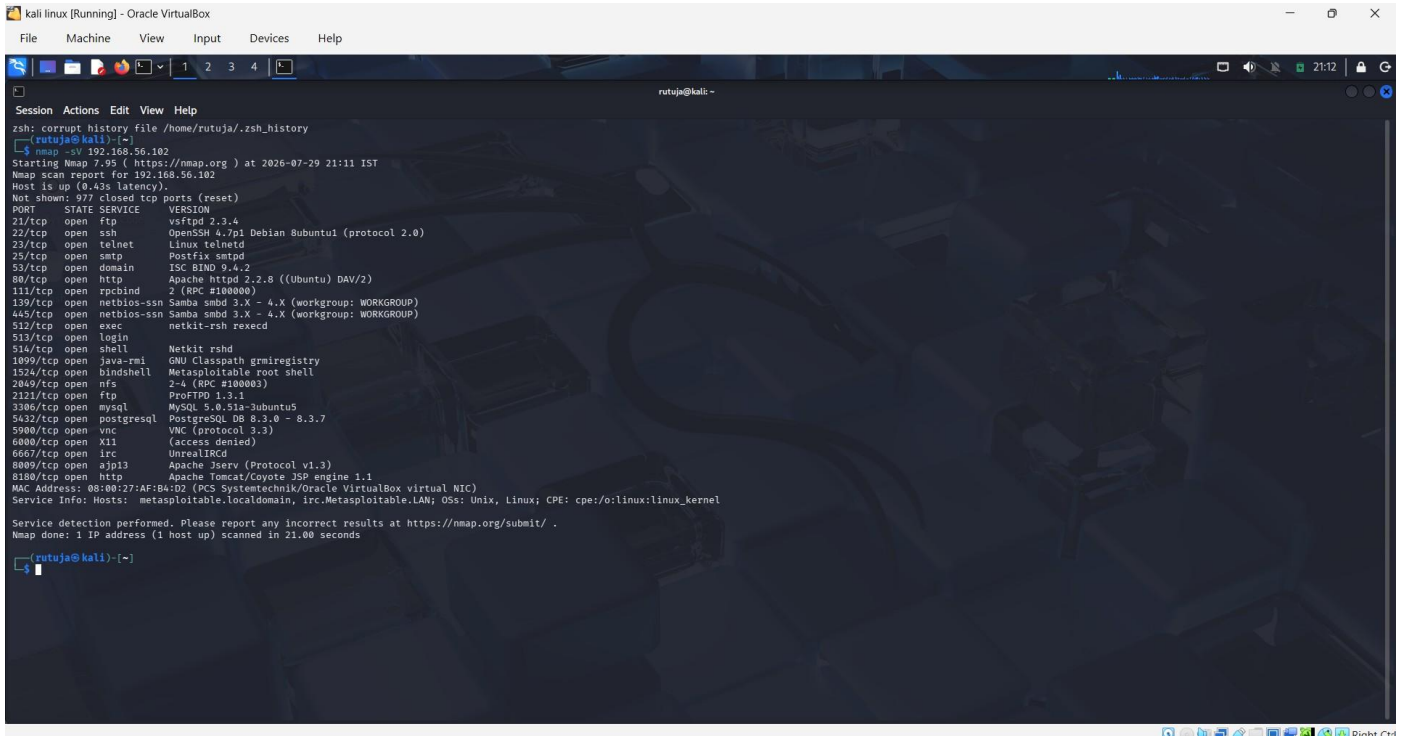
PORT      STATE      SERVICE
53/udp    open       domain
67/udp    closed     dhcp
68/udp    open|filtered dhcp
69/udp    open|filtered tftp
123/udp   closed     ntp
135/udp   closed     msrpc
137/udp   open       netbios-ns
138/udp   open|filtered netbios-dgm
139/udp   closed     netbios-ssn
161/udp   closed     snmp
162/udp   closed     snmptrap
445/udp   closed     microsoft-ds
500/udp   closed     isakmp
514/udp   closed     syslog
520/udp   closed     route
631/udp   closed     ipp
1434/udp  closed     ms-sql-m
1900/udp  closed     upnp
4500/udp  closed     nat-t-ike
49152/udp closed     unknown
MAC Address: 08:00:27:AF:B4:D2 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 16.90 seconds
rutuja@kali:~$
```

7. Service Version Detection (-sV)

Service version detection attempts to identify the application or network service listening on open ports and determine its version. This information is useful when assessing outdated or vulnerable software.

Command: nmap -sV 192.168.56.102



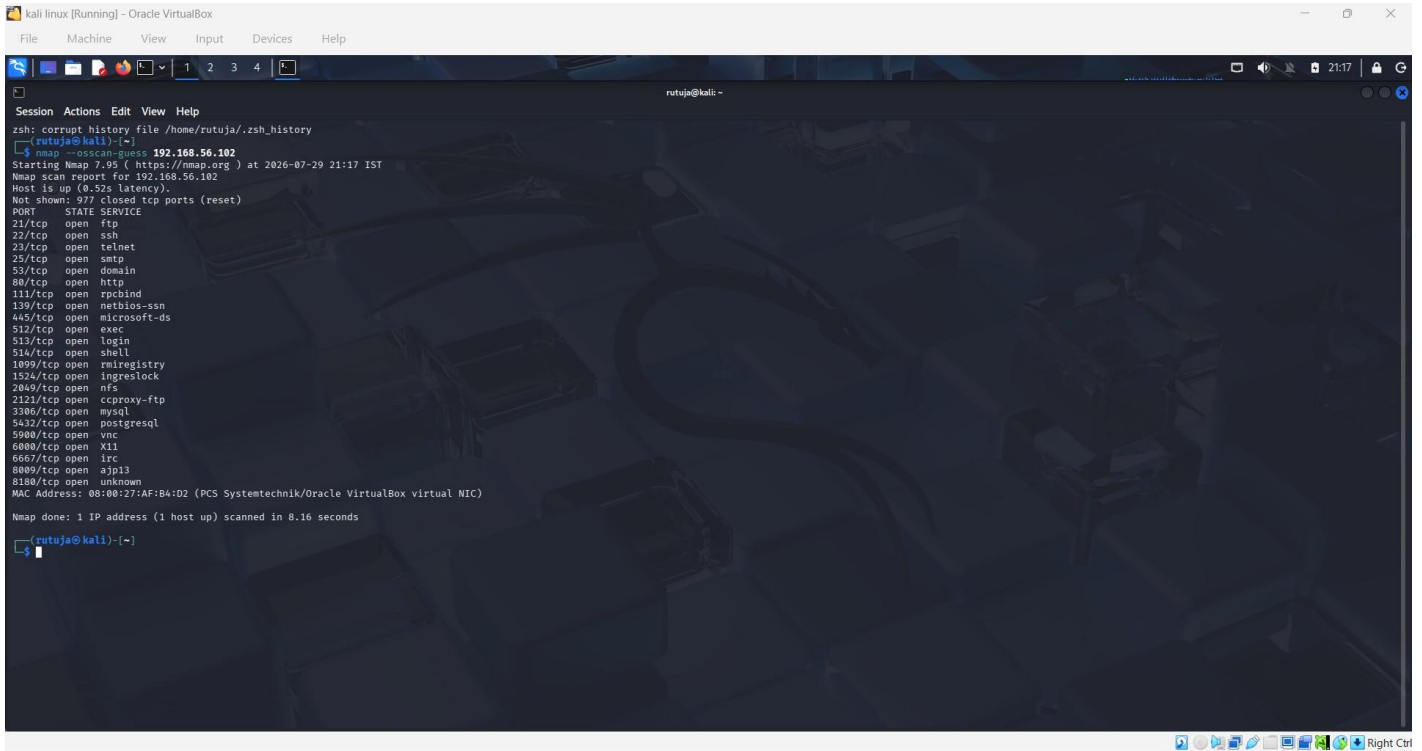
```
zsh: corrupt history file /home/rutuja/.zsh_history
(rutuja@kali) ~
$ nmap -sV 192.168.56.102
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-29 21:11 IST
Nmap scan report for 192.168.56.102
Host is up (0.43s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
21/tcp    open  ftp            vsftpd 2.3.4
22/tcp    open  ssh            OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet         Linux telnetd
25/tcp    open  smtp           Postfix smtpd
53/tcp    open  domain         ISC BIND 9.4.2
80/tcp    open  http           Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind        2 (RPC #100000)
139/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec           netkit-rsh rshcd
513/tcp   open  login          netkit-rsh rshcd
514/tcp   open  shell          Netkit rshd
18099/tcp open  java-rmi       GNU Classpath gcmregistry
1524/tcp  open  bindshell      Metasploitable root shell
2049/tcp  open  nfs            2-4 (RPC #100003)
2121/tcp  open  ftp            ProFTPD 1.3.1
3306/tcp  open  mysql          MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql     PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc            VNC (protocol 3.3)
6000/tcp  open  x11            (access denied)
6667/tcp  open  irc            UnrealIRCd
8009/tcp  open  ajp13          Apache Jserv (Protocol v1.3)
8180/tcp  open  http           Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:AF:B4:D2 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OS: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 21.00 seconds
(rutuja@kali) ~
```

8. OS Detection (-O)

Nmap OS detection uses network fingerprinting to estimate the target operating system. Results depend on the target's responses, network conditions, and scan privileges.

Command: `nmap -ossan-guess 192.168.56.102`



```
kali linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help

rutuja@kali: ~
Session Actions Edit View Help
zsh: corrupt history file /home/rutuja/.zsh_history
[rutuja@kali]~$ nmap -ossan-guess 192.168.56.102
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-29 21:17 IST
Nmap scan report for 192.168.56.102
Host is up (0.52s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmlregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8000/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:AF:B4:D2 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 8.16 seconds
[rutuja@kali]~$
```

9. Findings and Security Analysis

The final findings must be based on the actual Nmap output. Review the scan screenshots and record the open/filtered ports, detected services, versions, and OS information. Do not add ports or versions that were not actually observed.

Key points to analyze include unnecessary exposed services, insecure protocols, outdated service versions, remotely accessible administrative services, and services that should be restricted by firewall rules.

10. Recommendations

- Disable unnecessary services and close unused ports.
- Replace insecure remote-access protocols such as Telnet with SSH.
- Keep services and operating systems patched.
- Restrict administrative services to trusted hosts.
- Use firewall rules to allow only required traffic.
- Repeat Nmap scans after remediation to verify improvements.

11. Conclusion

The practical demonstrated TCP SYN scanning, UDP scanning, service/version detection, and OS detection using Nmap in a controlled lab. These techniques provide useful visibility into a target system and form an important part of a vulnerability-assessment process.